

cv_emma_andersson



Curriculum Vitae



Personal Information

Full Name	Emma Kristina Andersson
Preferred Name	Emma Andersson
Date of Birth	August 22, 1987
Gender	Female
Nationality	Swedish
Current Country of Residence	Sweden
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Academic Background



Education

Doctor of Philosophy (Ph.D.)

KTH Royal Institute of Technology

September 2013 - June 2018

- **Major:** Computer Science
- **Research Field:** Machine Learning & Long Sequence Modeling
- **Research Focus:**
 - Efficient attention mechanisms for long-range dependencies
 - Memory-augmented neural networks for sequence processing
 - Computational efficiency in transformer architectures
- **Advisor:** Prof. Lars Bergström

- **Thesis:** "Efficient Architectures for Long Sequence Processing in Neural Networks"

Master of Science (M.S.)

Uppsala University

September 2009 - June 2011

- **Major:** Computer Science
- **Research Field:** Artificial Intelligence
- **GPA:** 4.8/5.0
- **Thesis:** "Recurrent Neural Networks for Time Series Prediction"

Bachelor of Science (B.S.)

Lund University

September 2006 - June 2009

- **Major:** Computer Science
- **Minor:** Applied Mathematics
- **GPA:** 4.7/5.0 (Rank: 3/145)
- **Capstone Project:** "Pattern Recognition in Sequential Data"

Academic Honors & Awards

Year	Honor	Institution
2024	ICML 2024 Outstanding Paper Award	International Conference on Machine Learning
2023	Wallenberg AI, Autonomous Systems and Software Program (WASP) Distinguished Researcher Award	Knut and Alice Wallenberg Foundation
2022	Swedish Research Council Starting Grant	Vetenskapsrådet (Swedish Research Council)

Selected Publications

Journal Articles

1. **Andersson Emma**, Bergström Lars, Nielsen Henrik (2024). "Linear-Time Attention Mechanisms for Ultra-Long Sequence Modeling." *Nature Communications*, 15(1), 3456-3472. [IF: 16.6]

doi: [10.1038/s41467-024-45678-x](https://doi.org/10.1038/s41467-024-45678-x)

 **Citations: 187** |  Open Access

2. **Andersson Emma**, Johansson Maria, Bergström Lars (2023). "Memory-Efficient Transformers for Processing Million-Token Sequences." *Journal of Machine Learning Research*, 24(89), 1-48. [IF: 6.0]

doi: [10.5555/3648699.3648788](https://doi.org/10.5555/3648699.3648788)

 **Citations: 342**

Conference Papers

1. **Andersson Emma**, Chen Wei, Bergström Lars (2024). "Hierarchical State Space Models for Long-Context Language Understanding." In *Proceedings of the 41st International Conference on Machine Learning (ICML 2024)*, pp. 1234-1246. Vienna, Austria.

 **Outstanding Paper Award**

 **Citations: 156**

2. Olsson Per, **Andersson Emma**, Nielsen Henrik (2023). "Sparse Attention Patterns for Efficient Long Sequence Processing." In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, pp. 12345-12358. New Orleans, USA. [CCF A]

 **Citations: 289**

3. **Andersson Emma**, Lindström Sofia (2023). "Adaptive Context Windows for Variable-Length Sequence Modeling." In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL 2023)*, pp. 8765-8778. Toronto, Canada.

 **Citations: 421**

4. **Andersson Emma**, Bergström Lars, Wu Tianyi (2022). "Recurrent Memory Transformers for Long Document Understanding." In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing (EMNLP 2022)*, pp. 5432-5445. Abu Dhabi, UAE.

 **Citations: 534**

Preprints

1. **Andersson Emma**, Kumar Rajesh, Bergström Lars, et al. (2024). "Scaling Attention to Infinite Context: Theory and Practice." *arXiv preprint arXiv:2412.xxxxx*.

 Under review at ICLR 2025

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 **GitHub:** github.com/emmaan1dersson

 **Google Scholar:** scholar.google.com/citations?user=xxxxxx

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